

1 Product Introduction

The low-altitude surveillance radar is a detection device for UAV defense and control. It is a three-coordinate pulse-Doppler radar with mechanical scanning in azimuth and phased scanning in elevation, mainly used to detect and locate low-altitude aircraft. The radar can accurately detect the spatial position of targets and can be cascaded with jamming, navigation spoofing, and other countermeasure devices.



Figure 1 Low-Altitude Surveillance Radar

2 Key Features

- Adopts a solid-state active transceiver system design and digital signal processing technology, with high reliability, safety, supportability, testability, operability, and maintainability.
- Provides omnidirectional, all-weather, round-the-clock automatic target detection and tracking capability. Supports normal

operation at night and under rain, snow, dense fog, and other conditions.

- Uses weak-small target detection and recognition technology based on target motion characteristics, enabling effective warning, precise measurement, and data reporting for target UAVs.

- Provides detection and tracking capability for multiple batches of targets.

- Provides 360° azimuth search for blind-spot-free azimuth detection.
- Provides online monitoring, fault diagnosis, and fault location capabilities.

Basic Specifications	
Radar Band	X band (9 GHz-10.2 GHz)
Detection Range (False alarm rate: 10 ⁻⁶ ; detection probability: 85%)	Data rate 6 s: ≥10 km (DJI Phantom 4), ≥7 km (Egret) Data rate 3 s: ≥8 km (DJI Phantom 4), ≥5 km (Egret)
Detectable Target Speed	1-200 m/s
Near-Range Blind Zone	≤200 m
Circular Scanning Coverage	Azimuth: 0-360°; elevation: 0-30°
Target Capacity	≥200 targets
Accuracy	Azimuth: ≤0.5°; elevation: ≤0.5°; range: ≤10 m
Self-Test Capability	Real-time subsystem operating status monitoring
Communication Protocol	UDP
Total Power Consumption	≤400 W
Power Supply	AC 220 V / 50 Hz
Data Interface	Gigabit Ethernet port
Overall Dimensions	≤990 mm × 450 mm × 650 mm (L×W×H)
Total Weight	≤66 kg (excluding tripod and rain cover)